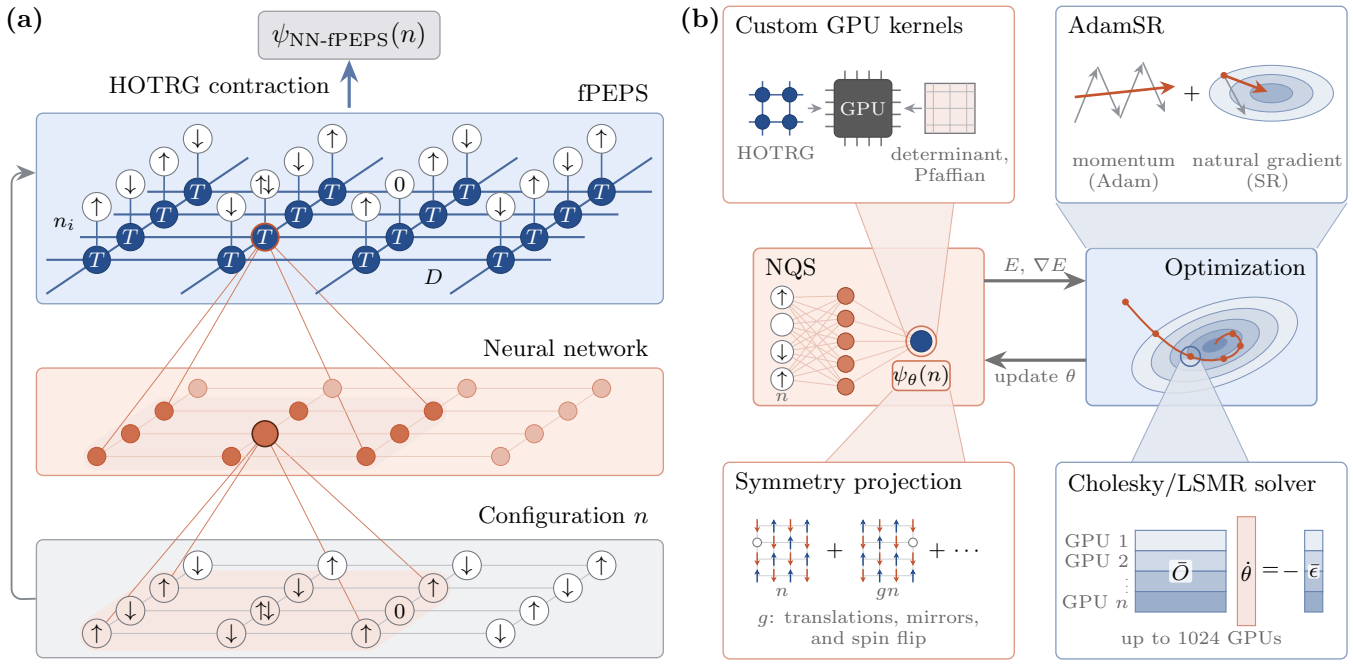

(a)
$\psi_{\text{NN-fPEPS}}(n)$
HOTRG contraction
fPEPS
$n_i$
$D$
Neural network
Configuration $n$
(b)
Custom GPU kernels
GPU
HOTRG
determinant, Pfaffian
AdamSR
momentum (Adam)
natural gradient (SR)
NQS
$\psi_\theta(n)$
$E$, $\nabla E$
update $\theta$
Optimization
Symmetry projection
$g$: translations, mirrors, and spin flip
Cholesky/LSMR solver
GPU 1
GPU 2
GPU $n$
$\bar{O}$
$\dot{\theta} = -\bar{\epsilon}$
up to 1024 GPUs